stm32-blinky — Design Document

ai-ee v1 pipeline

generated 2026-08-14 13:59:08

Contents

1 Board and Run Metadata

Field	Value
board	stm32-blinky
workspace	`boards/stm32-blinky`
phase	P10
gate status	all 5 recorded gates pass
state created	2026-07-27T16:26:45
state updated	2026-08-14T10:38:35

2 Overview

Brief: stm32-blinky (S14 run a)

A minimal 2-layer STM32 development blinker board:

- STM32F103C8T6 MCU
- 8 MHz crystal
- One user LED on a GPIO
- 4-pin SWD debug header
- Powered from a 2-pin 5V pin header through a 3.3V LDO (AMS1117-class)
- JLCPCB 2-layer, JLC economy PCBA
- Roughly 50x40 mm or smaller

- Prototype qty 5
- No USB, no battery
- Prefer JLC Basic parts throughout

Fabrication spec snapshot

Key	Value
layers	2
width_mm	50.0
height_mm	40.0
qty	5
surface_finish	HASL
solder_mask_color	green
assembly	True

Stackup

Chosen stackup: `JLC2313_1.6` (board class: 2-layer)

3 Requirements

Requirements: stm32-blinky

Source: `brief/brief.md` (S14 run a). No other attachments. Unstated low-risk items are marked `ASSUMED:` inline; design-changing unknowns are in section 9.

1. Function

A minimal 2-layer STM32 development "blinker" board: an STM32F103C8T6 MCU clocked from an 8 MHz crystal, driving one user LED from a GPIO. Programmed and debugged over a 4-pin SWD header. Powered from an external 5 V supply via a 2-pin header, regulated to 3.3 V by an AMS1117-class LDO. No USB, no battery, no radio. Firmware (the blink program) is out of scope for the hardware pipeline; the board must be flashable by the user over SWD.

2. Interfaces

```
| Interface | Details |
|---|---|
| 5 V power input | 2-pin pin header.  ASSUMED: 2.54 mm male header, polarity
marked on silkscreen.  |
| SWD debug | 4-pin header (SWDIO, SWCLK, 3V3, GND). ASSUMED: 2.54 mm single-row
male, ST-Link-compatible pin order.  Programming interface for the user; no
NRST pin on a 4-pin header (reset via SWD/power cycle).  |
| User LED | 1x LED on an MCU GPIO, with series resistor.  ASSUMED: any
standard-color LED available as a JLC Basic part; GPIO chosen at architecture
stage.  |
| None else | No USB, no RF, no buttons stated (see Q2 to confirm the omissions
are intentional).  |
```

3. Power

- Input: 5 V nominal on the 2-pin header. ASSUMED: pre-regulated external supply, 4.5-5.5 V; no wider input range required.

- Regulation: single 3.3 V rail from an AMS1117-class LDO (stated).
- Rail budget (GUESS, to be confirmed at architecture): MCU worst-case ~40-60 mA, LED ~2-10 mA, total under 100 mA on 3.3 V. AMS1117-class parts (~800 mA) give ample margin; dissipation at 5 V -> 3.3 V, 100 mA is ~0.17 W - trivial, no thermal concern.
- No battery, no charging (stated).
- Reverse-polarity protection on the unkeyed 2-pin input: not stated - see Q1.

4. Environment

Not stated. ASSUMED: indoor bench/prototype use, commercial ambient (0-70 C), no enclosure, no ingress/vibration/condensation requirements.

5. Size & mounting

- Outline: 50 x 40 mm maximum ("roughly 50x40 mm or smaller"); treat 50 x 40 as the hard limit and smaller as preferred.
- Height: no limit stated. ASSUMED: none (pin headers dominate height).
- Mounting holes: not stated - see Q2 (default: none).

6. Quantity & budget

- Build quantity: 5 prototypes (stated). ASSUMED: all 5 assembled, not bare-PCB-only.
- Target unit cost: not stated. ASSUMED: no hard cap; the stated economy PCBA + Basic-parts preference already expresses "minimize cost".

7. Assembly

- JLCPCB, 2-layer board, JLC economy PCBA (stated).
- ASSUMED: single-sided (top) SMT assembly, consistent with the economy service.
- ASSUMED: the two THT pin headers may be shipped loose / hand-soldered by the user if JLC economy through-hole assembly is unavailable or adds disproportionate cost for qty 5.
- Parts: prefer JLC Basic parts throughout (stated). ASSUMED: the explicitly specified STM32F103C8T6 overrides the Basic-parts preference if it is only stocked as an Extended part; the preference applies to all passives and support parts.

8. Compliance/safety flags

None apply: no mains, no battery, no motors, maximum voltage on board is 5 V (< 30 V), currents well under 3 A, no RF transmitter.

9. Open questions

1. Reverse-polarity protection: the 2-pin 5 V header is unkeyed and can be plugged in backwards, which would likely destroy the board. Add a protection diode? It adds one cheap Basic part and slightly reduces regulator headroom (still fine from a 5 V supply at this load). Default: yes, add it.
2. The brief reads as deliberately minimal. Confirm that NONE of the following commonly-added extras are wanted: reset button, BOOT0/bootloader jumper, power-indicator LED, mounting holes. Default: none of them (keep the board minimal as briefed). If you do want any, name them.

Answers (P0 batch, user-approved 2026-07-27)

1. Reverse-polarity protection: YES - add a series protection element (Schottky diode or P-FET, JLC Basic part) on the 5V input before the LDO. 2. Extras: NONE - no reset button, no BOOT0 jumper, no power LED, no mounting holes. Board stays as briefed. Checkpoint policy note: H3 (placement render) folds into H4 for this run.

4 Architecture

Decisions of record (state.json)

Phase	What	Why
P0	Add reverse-polarity protection on the 5V input (series diode/P-FET, Basic part)	User-approved P0 answer: unkeyed 2-pin header can be reversed
P0	Keep board minimal: no reset button, no BOOT0 jumper, no power LED, no mounting holes	User-approved P0 answer: brief is deliberately minimal
P0	Checkpoint H3 folded into H4 (placement render attached to verification checkpoint)	User-approved: keeps run at the spec M5 <=5 human interactions bar
P1	Skip all P1 researchers (component-scout, reference-design, interface-spec, power-architect)	Trivially powered (5V header->LDO->3.3V <100mA), no novel blocks, no standards-bound interfaces; parts named in brief; P3 part-sourcer covers sourcing
P2	Stackup JLC2313_1.6, 2-layer, target 35x30mm, B.Cu GND pour	Blinky-class density needs no inner planes; JLC economy 2L
P2	Rev-polarity = series SS34 Schottky (Basic); AO3401A P-FET is the named escalation if headroom concerns win at H1	One Basic part; headroom closes at real ~40mA load (typ-only at 100mA abuse case)
P2	Single flat root sheet; refdes plan U1/U2/J1/J2/D1/D2/Y1/R1/R2/C1-C9; LED on PC13 active-low via 1k	Board size does not warrant hierarchy; PC13 sink limited to 3mA per DS5319
P3	Crystal = C12674 YXC HC-49S-SMD 20pF Basic (architect's from-memory LCSC C32828 was an ADI vref - corrected by live search)	parts_search is the only source of part-number truth
P3	Headers C32713268/C32713270 vertical THT, hand-solder (economy PCBA is SMT-only); board SMT run = 9 Basic + STM32 Extended	No Basic pin headers exist on JLC
P3	User directive: all remaining checkpoints/decisions auto-approved by orchestrator with recommended defaults, for ALL S14 runs; report only at S14 completion	User message 2026-07-27: 'Going forward just make the decisions yourself and tell me when s14 is done'
P5	S14 exec plan: model-mix (sonnet/opus subagents, fable reviewers only, script phases inline); run (b) stays full	User-approved cost control 2026-07-27

Phase	What	Why
P5	Outline fixed 50x40 (requirements cap) at P5; 35x30 architecture target unreachable (no outline-shrink step exists, v1 limit). LED courtyard-smaller-than-pads quirk deferred unless P6/P7 trip	Cap must bind at P5; shelf-pack needs the room
P7	Take P7->P6 backward edge: restore pre-route snapshot, ~1mm west-face relief at U1 per router analysis, re-route	U1.8 GND fan-out sealed by /NRST via + +3V3 escape; only legal via site had 0.009mm margin (router refused coin-flip); stitch no_clear_spot + KRT finish agree
P8	verify fail triage: pdn_undecoupled /VIN = width-only stub -> new 'pdn:false' opt-out (check_pdn+schema); pdn_no_bulk +5V/+3V3 = parse_farads multi-token bug -> fixed with tests	Both check-side defects, board is correct; S14 hardening mandate
P9	dfm a1 triage: 12 silk-sliver errors (0.003-0.02mm2) = check-side threshold gap -> dfm_check warn band <0.05mm2 added w/ tests; 2 copper-clearance 0.035mm same-net sliver gaps -> router point-fix; bom missing_lcsc = bom_cpl now reads board LCSC fields (S14 fix) w/ tests	Board is KiCad-clean; gerber-side geometry + mapping gaps were script defects

Sheet plan

stm32-blinky - sheet plan

One flat root sheet

Single root sheet (`stm32-blinky.kicad_sch`), no hierarchy. Justification: 18 components / 6 blocks fit one readable page; hierarchical sheets would add sheet-pin/label drift risk (LEARNINGS 2026-07-22: child nets rename to "/<sheet>/NAME", root-side names win at pins) for zero navigational gain. The proven blinky2 golden is flat. Placement grouping at P6 comes from constraints.json + decoupling.json, not from sheets, so hierarchy buys nothing there either.

Consequence for net names: power nets are bare (`+5V`, `+3V3`, `GND`), every labeled signal net is a root local label and lands in the netlist as `/NAME`. The names below are CANONICAL - P4 must produce exactly these and constraints.json already references them.

Refdes plan (canonical assignment - single sheet, one range)

```
| Refdes | Part | Block |
|---|---|---|
| U1 | STM32F103C8T6, LQFP-48 | MCU core |
| U2 | AMS1117-3.3, SOT-223 | Regulation |
| J1 | 1x2 2.54 mm header (5 V in) | Power input |
| J2 | 1x4 2.54 mm header (SWD) | Debug |
| D1 | SS34 Schottky, SMA | Reverse protection |
| D2 | Red LED 0603/0805 | User LED |
| Y1 | 8 MHz crystal, HC-49S SMD | Clock |
| R1 | 1 k 0603 (LED series) | User LED |
| R2 | 10 k 0603 (BOOT0 pulldown) | MCU core |
| C1-C3 | 100 nF 0603 (VDD 48/24/36) | MCU core |
| C4 | 100 nF 0603 (VDDA pin 9) | MCU core |
```

C5	10 uF 0805 (LDO out)	Regulation
C6	10 uF 0805 (LDO in)	Regulation
C7, C8	22 pF C0G 0603 (xtal load)	Clock
C9	100 nF 0603 (NRST)	MCU core

Power-symbol refs: `pwr_base=1` (#PWR001 upward, single range - no other sheet to collide with).

constraints.json references J1, J2 (edges) and Y1, C7, C8 (xtal group); decoupling.json (P4-emitted) will reference C1-C6 against U1/U2 pins as in the table above. These assignments are therefore load-bearing, not suggestions.

Canonical nets

Net	From -> to	Notes
`/VIN`	J1.1 -> D1 anode	raw unprotected 5 V stub; no caps here
`+5V`	D1 cathode -> C6, U2.VIN(3)	protected rail; needs PWR_FLAG (diode-fed)
`+3V3`	U2.VOUT(2/tab) -> C5, C1-C4, U1 VDD/VDDA/VBAT, R1, J2.3	driven by LDO power output
`GND`	J1.2, U2.GND(1), all cap returns, R2, J2.4, Y1 can (if 4-pad)	needs PWR_FLAG; B.Cu pour at P7
`/SWDIO`	U1 PA13 (pin 34) -> J2.1	
`/SWCLK`	U1 PA14 (pin 37) -> J2.2	
`/LED_A`	R1 -> D2 anode	R1 top side ties to +3V3
`/LED`	D2 cathode -> U1 PC13 (pin 2)	ACTIVE-LOW, PC13 sinks ~1.3 mA
`/OSC_IN`	U1 PD0/OSC_IN (pin 5) -> Y1, C7	high_speed w/ GND reference
`/OSC_OUT`	U1 PD1/OSC_OUT (pin 6) -> Y1, C8	high_speed w/ GND reference
`/NRST`	U1 NRST (pin 7) -> C9	no header pin, no button (P0 answer)
`/BOOT0`	U1 BOOT0 (pin 44) -> R2 -> GND	fixed strap, not a jumper

U1 pin commitments (P4 wiring contract, LQFP-48)

- VBAT (1) -> +3V3 (no battery). VDD: 24/36/48, VDDA: 9 -> +3V3. VSS: 23/35/47, VSSA: 8 -> GND.
- PC13 (2) = /LED sink. PD0 (5)//OSC_IN, PD1 (6)//OSC_OUT. NRST (7) = /NRST.
- PA13 (34) = /SWDIO, PA14 (37) = /SWCLK. BOOT0 (44) = /BOOT0.
- PB2/BOOT1 (20): floating is electrically fine (don't-care with BOOT0=0) but give it a no-connect flag like every other unused GPIO so ERC is clean.

Interface pinouts (silk-labeled, canonical)

- J1: 1 = +5 V in (/VIN), 2 = GND. Polarity marked on silk (requirements).
- J2: 1 = SWDIO, 2 = SWCLK, 3 = 3V3, 4 = GND (requirements' listed order).

Board-edge placement (mirrors constraints.json)

- J1 on the LEFT edge (power enters beside the D1 -> U2 chain; U1's crystal/LED pins face left-top too, keeping the left half analog-quiet-ish).
- J2 on the RIGHT edge (PA13/PA14 sit on the LQFP right/top-right flank).

Full architecture narrative (not inlined; see the artifact index): `architecture/blocks.md`, `architecture/decisions.md`, `architecture/power_tree.md`, `architecture/stackup.md`.

5 Schematic

Gate `erc` (phase P4): **pass** after 3 attempt(s); last 2026-08-14T10:05:48, failing 0 of 0.

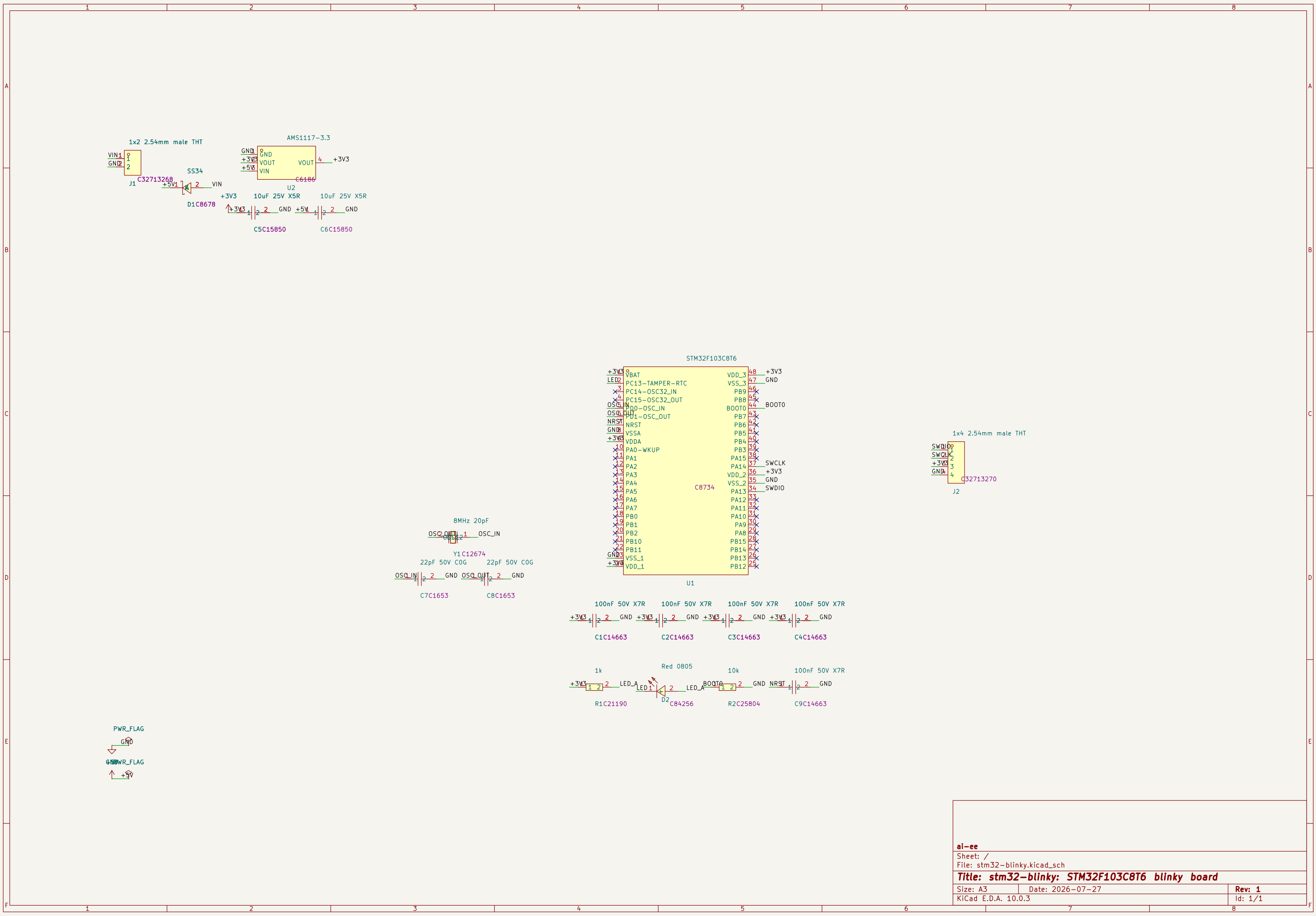
Review waivers

ERC/schematic review waivers (stm32-blinky)

W1 (reviewer warning 4, issue 4): "erc-pintypes-vacuous" - easyeda2kicad symbols carry mostly-passive pin types even after datasheet-driven retyping of supplies, so ERC's driver-conflict/float checks have reduced power on this board. Waived: connectivity was independently hand-verified by the adversarial reviewer against both datasheet extracts (all 9 supply pins, polarity, strapping). Tooling fix (retype at lib_pull time, GPIOs as bidirectional) queued as S14 hardening item.

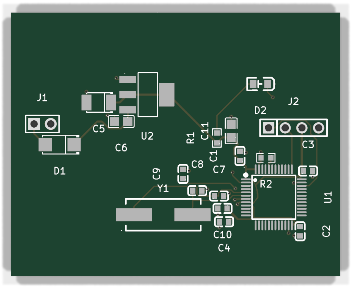
Note: reviewer warnings 2 (VDDA 1uF//10nF) and 3 (VDD_3 bulk) were FIXED, not waived, along with error 1 (22uF tantalum LDO output cap) - see state.json issues 1-3 and the second erc gate commit.

The full schematic PDF follows.

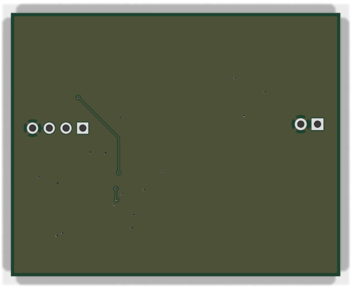

1x2 2.54mm male THT
J1
C32713268
AMS1117-3.3
U2
C6186
SS34
D1 C8678
10uF 25V X5R
C5 C15850
C6 C15850
STM32F103C8T6
U1
C8734
1x4 2.54mm male THT
J2
C32713270
8MHz 20pF
Y1 C12674
22pF 50V C0G
C7 C1653
C8 C1653
100nF 50V X7R
C1 C14663
C2 C14663
C3 C14663
C4 C14663
C9 C14663
1k
R1 C21190
Red 0805
D2
C84256
10k
R2 C25804
PWR_FLAG
ai-ee
Sheet: /
File: stm32-blinky.kicad_sch
Title: stm32-blinky: STM32F103C8T6 blinky board
Size: A3
Date: 2026-07-27
KiCad E.D.A. 10.0.3
Rev: 1
Id: 1/1

6 Layout

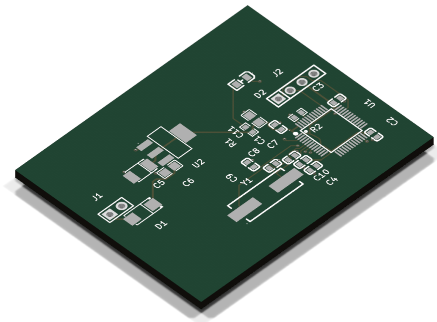
Gate `place` (phase P6): **pass** after 2 attempt(s); last 2026-08-14T10:38:08, failing 0 of 0.
Gate `drc_routed` (phase P8): **pass** after 4 attempt(s); last 2026-08-14T10:38:31, failing 0 of 0.



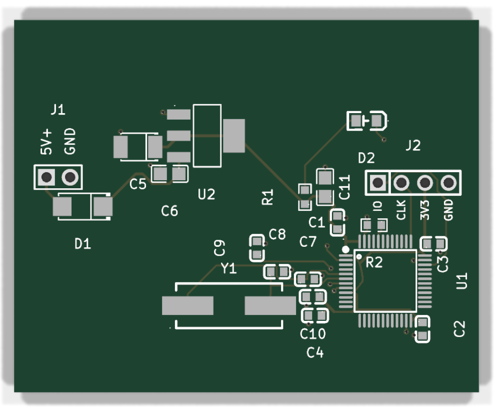
`reports/renders/stm32-blinky_top.png`



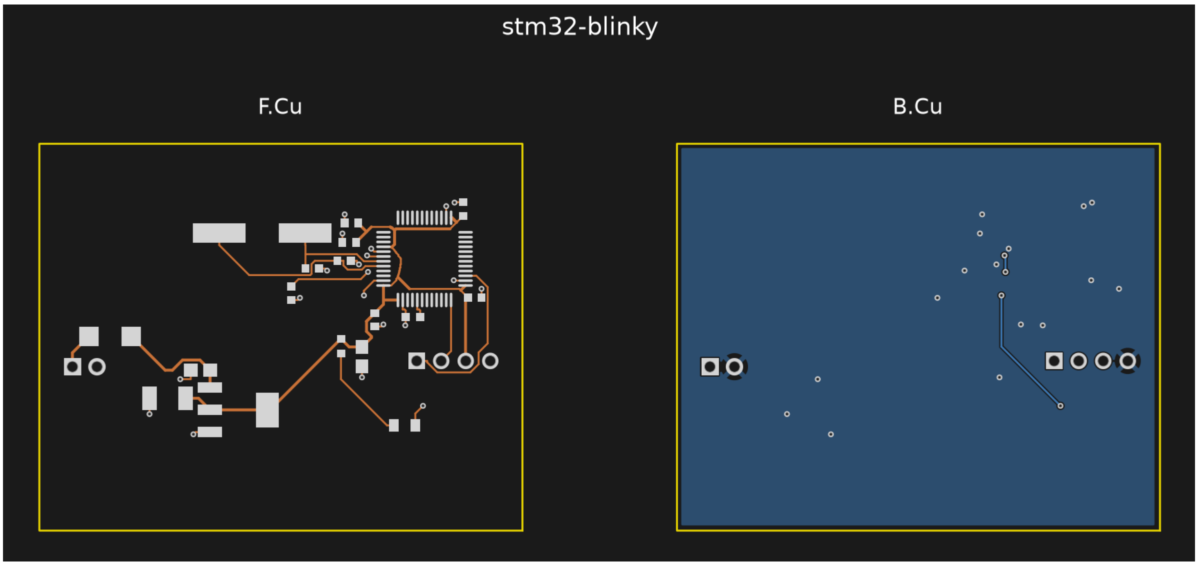
`reports/renders/stm32-blinky_bottom.png`



`reports/renders/stm32-blinky_iso.png`



`reports/render_labeled/stm32-blinky_top.png`



`reports/layers/stm32-blinky_layers.png`

7 Verification

Gate `verify` (phase P8): **pass** after 4 attempt(s); last 2026-08-14T10:38:32, failing 0 of 5.

Verification checks

Check	Status	Findings	Note
`check_return_path`	pass	0	
`check_current`	pass	0	
`check_decoupling`	pass	0	
`check_diffpair`	pass	0	
`check_creepage`	pass	0	
`check_thermal`	pass	0	
`check_silk`	pass	0	
`check_pdn`	pass	0	

Design review of record

Board review: stm32-blinky (P8 adversarial pass)

Reviewer: verify-reviewer (fresh context). Machine gate state re-verified independently: verify_all re-run from scratch on the current board = PASS, all 8 checks 0 violations (return_path, current, decoupling, diffpair, creepage, thermal, silk, pdn). DRC routed 0/0. What follows is what the scripts cannot see.

Renders referenced (made this pass, reports/renders/): `stm32-blinky_top.png`, `stm32-blinky_bottom.png`, `stm32-blinky_iso.png`, crops `crop_j1_d1.png`, `crop_c5_tant.png`, `crop_d2_j2.png`, `crop_u1_cluster.png`, `crop_xtal.png`.

—

ERROR 1 - J1 power input has no polarity legend on silk (requirements miss)

`requirements.md` sec 2 and `architecture/blocks.md` both promise the 5 V input "polarity marked on silkscreen" (J1 pin 1 = +5 V, pin 2 = GND). The as-built board has NO text or +/- legend anywhere: silk carries refdes only (there is not a single gr_text on the board). What exists at J1 is a pin-1 dot and a square pad (see `crop_j1_d1.png`) - and the dot sits on the header body outline, so after the header is soldered the plastic covers it. Post assembly the top side shows nothing but "J1"; the square pad is visible only from the bottom.

Consequence: the unkeyed 2-pin input can be plugged backwards with no on-board hint. D1 (series SS34) prevents damage - reversed input just does not conduct - but the user gets a dead board and no way to know why or which way is right without opening the design files. This is the board's primary human interface and an explicit requirements promise. Fix is two silk texts ("5V", "GND" or "+", "-") next to the holes, outside the header body. Must fix before fab; there is ample empty silk area (left region is empty).

WARNING 1 - J2 SWD pins unlabeled (architecture promise miss)

`blocks.md` promises "Canonical pin order (silk-labeled): 1 = SWDIO, 2 = SWCLK, 3 = 3V3, 4 = GND". Electrical order verified correct from the netlist (pads: SWDIO/SWCLK/+3V3/GND, U1 PA13/PA14 to pins 1/2). But there are no labels - only a pin-1 square cell (`crop_d2_j2.png`) which the header body covers after assembly. ST-Link clones are wired pin-by-pin; swapping 3V3/GND shorts the debugger's 3.3 V rail to ground. Same trivial fix as ERROR 1 (four small silk texts below the header row - the area under J2 is clear). Strongly recommend fixing together with ERROR 1; warning rather than error only because requirements.md itself demands the ORDER (met), the labels are an architecture-level promise.

WARNING 2 - refdes association ambiguity (cap cluster + C3-under-J2)

Confirmed the P6-flagged issue on `crop_u1_cluster.png` and `crop_d2_j2.png`; judged as follows:

- "C11" text sits directly above R1's body; "C1" text below-left of R1; "C7"/"C8" labels float between parts they do not touch. Among these 0603s are four DIFFERENT cap values (22 pF, 10 nF, 100 nF, 1 uF) plus a 1 k resistor - a hand-rework or debug probe guided by silk has a real chance of landing on the wrong part.
- "C3" text renders directly under J2 pads 3-4 and reads like a header label (its own body is 3.6 mm away). Mildly reinforces the J2 labeling problem: someone may read "C3" while counting SWD pins.

Assembly itself is CPL-driven, so production risk is nil; consequence is limited to bench rework/debug confusion. WAIVER RECOMMENDED for rev A - but since ERROR 1 forces a silk edit session anyway, nudging these labels onto their parts at the same time costs minutes.

WARNING 3 - BOM of record drifted from the board (100 nF qty, stale roles)

`parts/parts.json` (LCSC BOM of record) lists C14663 100 nF at `qty_per_board: 5` with role "C1-C3 VDD + C4 VDDA 100nF, C9 NRST (5/board)". The board mounts FOUR 100 nF (C1, C2, C3, C9): the schematic-review fix that added the DS5319 VDDA pair made C4 = 1 uF (C15849) and C10 = 10 nF (C57112), and the 100 nF line was never decremented. The 1 uF / 10 nF entries carry no refdes in their roles. Anything in P9 that quotes or orders from parts.json over-counts one part line and cannot map refdes for two others. One-line fix before ordering. (Values/nets on the board itself verified correct: VDDA pin 9 gets 1 uF + 10 nF, three VDD pairs get 100 nF, NRST gets 100 nF.)

WARNING 4 - stale reports/verify_all.json contradicts the gate

`reports/verify_all.json` (written 00:31, before commit e827d25's check_pdn parse fix + /VIN width-only opt-out) still records 1 error / 2 warnings, while `reports/gate-verify.json` says pass 0/0. An auditor at checkpoint 4 reading the named summary artifact sees a failing state that no longer exists. Re-running verify_all (done this review, to scratchpad) yields PASS with 0 violations - refresh the committed artifact.

—

Warnings triage (every prior verify_all finding gets a verdict)

The 00:31 verify_all's three findings, against the current board:

1. ERROR "power rail /VIN has no decoupling capacitors" - JUSTIFIED OPT-OUT (now `"pdn": false` in constraints.json). /VIN is the 4 mm raw stub from J1.1 to D1's anode. A capacitor there would sit in front of the reverse-protection diode: useless for the LDO (C6 10 uF sits on the protected +5V node 1.9 mm from U2's input, verified) and wrong for a polarized part (it would see reverse voltage in the fault case the diode exists for). Endorse the opt-out as permanently correct, not a convenience waiver. 2. WARNING "+5V ... no bulk reservoir; only ceramics" - PARSER ARTIFACT, RESOLVED. Old check_pdn could not parse "10uF 25V X5R"-style values (its own report said total_uf 0.0 with caps present). Post-fix run: +5V bulk_count 1, 10.0 uF total (C6). No board change was needed or made. 3. WARNING "+3V3 ... no bulk reservoir" - same artifact, RESOLVED. Post-fix: 7 caps, bulk_count 3, 33.3 uF total (C5 = 22 uF tantalum, ESR-matched to the AMS1117 per the settled user decision; C11 = 10 uF; C2/C3/C1 100 nF; VDDA pair). Real bulk was always present.

Fresh run has zero warnings and zero skipped checks: diff_pairs is an explicit empty list (no pairs exist - correct), thermal no-ops by documented budget (77 mW worst on U2, SOT-223 - trivially fine), creepage trivial at 5 V. No skipped-check holes in the verify suite. DFM/fab checks (dfm_check, fab export) are P9 scope and have NOT run yet - `fab/` is empty; that hole belongs to the next phase, not this gate.

Hunted and found clean (visual + cross-artifact)

- Polarity marks, decoded from silk geometry and checked against nets, all CORRECT and all visible after assembly: D1 cathode band + end brackets on the +5V side (brackets outside the SMA body); C5 tantalum "+" brackets on the pad-1/+3V3 end, outside the CASE-B body; D2 LED cathode chevron+bar on the /LED (PC13) end, outside the 0805 body (`crop_j1_d1.png`, `crop_c5_tant.png`, `crop_d2_j2.png`). U1 pin-1 dot clear of the LQFP body.

- Netlist vs intent: AMS1117 pinout (GND/VOUT/VIN/tab=VOUT) correct; D1 in series with cathode toward the LDO (conducts forward, blocks reverse); LED chain +3V3 -> R1 1k -> anode, cathode -> PC13 (active-low sink, ~1.3 mA, honors the PC13 3 mA / no-source limit); BOOT0 via R2 10k to GND; VBAT tied to +3V3; NRST 100 nF; PB2/BOOT1 floating (don't-care with BOOT0=0); all 3 VDD pairs + VDDA/VSSA mapped; SWD order per spec. Every requirements interface (5V in, SWD, LED) and every architecture block is present. No reset button / BOOT0 jumper / power LED / mounting holes - matches the user-confirmed minimal set.

- Connector reality: J1 3.5 mm from the left edge, J2 within 3.3 mm of the right edge, both vertical THT male headers with open space around and below - hand-solder access trivial (bottom face is bare pour, `stm32-blinky_bottom.png`), jumper access unobstructed, no tall parts anywhere near. Board is enclosureless by requirement, so no reach issues.

- EMI/robustness: no RF, no switcher on board (antenna/keepout hunt: N/A). Crystal Y1 is mid-board, 6.9 mm from the nearest edge, over continuous B.Cu GND pour; return-path check green. One deliberate oddity reviewed: C7 (OSC_IN load cap) sits at the MCU end, 12.6 mm from Y1 pad 1 (`crop_xtal.png`) while C8 hugs Y1 pad 2 - unconventional, but the 8 MHz Pierce loop rides a solid pour, the HC-49S itself is 11.5 mm long, and F103 HSE drive has wide margin. ACCEPT as-is (no board change); noting it so nobody "fixes" it into a worse route later.

- Bottom copper: near-solid GND pour, exactly one ~12 mm diagonal signal excursion plus one stub in the J2/D2 quadrant - nowhere near the oscillator or the LDO loop; stitching vias distributed. Clean.

- Assembly: single-sided top SMT (economy PCBA compatible), two THT headers hand-solder per settled plan. No fiducials - acceptable: JLC economy panelizes small boards on their own rails with their own fiducials; note only.
- Settled items verified as-built and NOT re-litigated: SS34 drop/headroom math (power_tree), 22 uF tantalum on LDO output present with correct polarity, minimal feature set, empty bottom-left region (intentional spread on the fixed 50x40 outline). One doc nit: blocks.md/decisions.md still say "target outline 35 x 30 mm"; the delivered board is the full 50 x 40 (within the hard limit, zero JLC cost impact at qty 5). Update the doc line or accept the drift - no board action.

Waiver recommendations (human decides at checkpoint 4)

- WARNING 2 (refdes ambiguity): waive for rev A if the silk session from ERROR 1 is not extended to labels - rework-only consequence.
- /VIN pdn:false opt-out: endorse permanently (see triage 1) - it is the electrically correct treatment, not a suppression.
- Architecture outline text 35x30 vs built 50x40: accept or one-line doc fix.

NOT waivable in this reviewer's judgment: ERROR 1 (J1 polarity legend) - explicit requirements promise, primary user interface, zero-cost fix.

Open (could not judge from renders alone)

- No 3D models render in the iso view (bare footprints), so assembled-body absurdities (header plastic orientation, crystal can height vs nothing) could not be visually confirmed; at this part mix the residual risk is negligible.
- fp_verify overlay reports exist for 6 footprints but were skipped for D1 (SS34/SMA) and Y1 (HC-49S). Mitigated here by manual geometry decode (D1 band-vs-net correct; Y1 pad span LS 12.7 matches HC-49S-SMD), but the physical-part-vs-footprint pad-1 convention for D1 rests on the LCSC footprint being band-on-pad-1, which lib_pull pulled but no overlay confirmed.

8 DFM and Fabrication

Gate `dfm` (phase P9): **pass** after 3 attempt(s); last 2026-08-14T10:38:35, failing 0 of 13. DFM findings by severity: warning: 13.

Fabrication outputs

Exported layers: F.Cu, B.Cu, F.Silkscreen, B.Silkscreen, F.Mask, B.Mask, F.Paste, B.Paste, Edge.Cuts.

Bill of materials

Comment	Designator	Footprint	LCSC
100nF 50V X7R	C1,C2,C3,C9	C0603	C14663
1uF 50V X5R	C4	C0603	C15849
22uF 16V tantalum	C5	CASE-B_3528	C8020
10uF 25V X5R	C6,C11	C0805	C15850
22pF 50V C0G	C7,C8	C0603	C1653
10nF 50V X7R	C10	C0603	C57112

Comment	Designator	Footprint	LCSC
SS34	D1	SMA_L4.3-W2.6-LS5.2-RD	C8678
Red 0805	D2	LED0805-RD	C84256
1x2 2.54mm male THT	J1	HDR-TH_2P-P2.54-V-M-3	C32713268
1x4 2.54mm male THT	J2	HDR-TH_4P-P2.54-V-M	C32713270
1k	R1	R0603	C21190
10k	R2	R0603	C25804
STM32F103C8T6	U1	LQFP-48_L7.0-W7.0-P0.50-LS9.0-BL	C8734
AMS1117-3.3	U2	SOT-223-3_L6.5-W3.4-P2.30-LS7.0-BR	C6186
8MHz 20pF	Y1	CRYSTAL-SMD_L11.5-W4.8-LS12.7	C12674

2 CPL rotation correction(s) applied; no missing LCSC numbers.

Quote

Estimated total $18.77 for qty 5 (unit $3.75). This is an estimate from published pricing; the JLCPCB cart quote page is authoritative.

Human steps before ordering

1. Upload boards\stm32-blinky\fab\stm32-blinky_gerbers.zip to https://jlcdfm.com/ and review the DFM report (V6: no public API).
2. Upload the same zip at https://cart.jlcpcb.com/quote and confirm the real price against the estimate above.
3. If assembling: upload BOM.csv and CPL.csv, then CHECK THE RENDERED PART PREVIEW for every polarized part (LED/diode/electrolytic) - rotation corrections are the classic failure mode.
4. Review, then pay. Payment is always the human's action.

9 Run Record (Appendix)

Phase digests

P0

```
# P0 digest (stm32-blinky)
Brief -> requirements.md.  Near-complete brief; 2 OPEN questions asked+answered
in one batch (interaction 1/5):  rev-polarity diode YES, extras NONE.
No safety flags (all <30V, <3A, no battery/mains/RF). Outline <=50x40mm,
qty 5, JLC economy PCBA, Basic parts preferred, 2-layer.
H3 folds into H4 for this run (user-approved).
```

P1

```
# P1 digest (stm32-blinky)
Research roster:  EMPTY by decision.  Trivially powered board, no novel blocks,
no standards-bound interfaces (SWD is a pinout convention).  The brief names
the MCU (STM32F103C8T6) and LDO class (AMS1117-3.3).  Sourcing happens at P3.
```

P2

```
# P2 digest (stm32-blinky)
6 blocks:  J1 5V -> D1 SS34 Schottky -> U2 AMS1117-3.3 -> U1 STM32F103C8T6
(LQFP-48) + Y1 8MHz crystal + D2 LED on PC13 (1k, 1.3mA sink) + J2 SWD.
Stackup JLC2313_1.6 2L, target 35x30mm, B.Cu GND pour.  Single flat sheet.
J1 left edge, J2 right edge (constraints placement.edges).  Nets canonical.
Cost ~$35-50 for 5 assembled; STM32 is the one Extended part.
Risk:  Schottky headroom at 4.5V low-line+100mA abuse (fine at real ~40mA);
P-FET AO3401A is the drop-in escalation.
```

P3

```
# P3 digest (stm32-blinky)
parts.json:  12 distinct parts, 9 Basic + STM32 Extended + 2 THT headers
(hand-solder; economy PCBA is SMT-only).  $2.77/board at qty 1 prices.
Architect's from-memory crystal LCSC corrected (C32828 vref -> C12674).
Extracts:  C8734 (48 pins x2-source cross-check, VDD 24/36/48 VDDA 9,
decoupling
100nF/pin + 4.7uF@VDD_3 + VDDA 1uF//10nF + NRST 100nF, LQFP48 land Fig63),
C6186 (pin1 GND/2 VOUT/3 VIN/tab VOUT, 22uF tantalum COUT stability).
Librarian:  12/12 pulled+registered, fp_verify 4 pass/0 fail/1 benign warning
(STM32 pad 0.27x1.50 vs 0.30x1.20 rec), courtyard-less list EMPTY, polarity
marks present+correct (SS34 band@K, LED chevron@pad1-cathode).  Tab=VOUT ok.
```

P4

```
# P4 digest (stm32-blinky)
Generator kicad/gen/root.py -> stm32-blinky.kicad_sch, ERC 0 err/0 warn
(severity-all), netlist_audit pass (20 components, 44 nets, 8/8 decoupling
associations, 0 value drift).  Adversarial review:  1 error + 3 warnings.
Fixed pre-board:  C5 -> 22uF tantalum C8020 (AMS1117 stability, anode on +3V3
verified), VDDA -> 1uF//10nF (C4/C10), VDD_3 bulk 10uF added (C11).
Waived:  pintypes-vacuous (erc-waivers.md; hardening item).  H2 AUTO-approved.
Refdes grew to C11 (sheets.md C1-C9 table stale - P2 doc drift, noted).
Interventions this phase:  lib pin-type retype (kicad/gen/lib_pin_types.py).
```

P5

```
# P5 digest (stm32-blinky)
board_init (inline, post-fix):  20 parts, 44 nets, 2L JLC2313_1.6, outline
FIXED 50x40 (requirements cap; margin 3), 0 mounting holes.  Self-check PASS:
parity 0 (LCSC field-copy fix live), setup 0 (D2 lib silk repaired by
librarian; 11 cross-part refdes silk marked transient - placement's to fix),
46 unconnected expected.  rules_gen:  2layer_1oz floors + 3 power width rules
(/VIN,+5V,+3V3 0.3A) + Power class; DRU beside board.  Sidecars in place.
Script fixes this phase:  parse_netlist fields + SetField/hide (parity bug),
samefile schematic skip, transient-silk partition.  28 tests green.
```

P6

```
# P6 digest (stm32-blinky)
seed 443->313 HPWL, anneal 3 candidates (best 160), chose cand1 + DRC-semantics
clearance repair + judgment edits (D2 out of U1.48 corridor; C1 8.7->2.7mm).
Final:  gate place PASS 0/0, HPWL 227, route probe 1.0 (46/46 at 25 passes),
KiCad DRC clean except 46 expected unconnected.  Crystal 4.1mm, decouplers
1.5-7.6mm all in class.  Renders reports/place/render_final/.
FINDING (major):  place gate passed 9 shorting pad pairs - courtyard-only
legality blind where EasyEDA courtyards < pad fields (LQFP48, D2).  Agent
built DRC-semantics repair (reports/place/tools/).  Hardening:  placelib
effective-courtyard = union(courtyard, pad bbox+margin) + fp_verify
courtyard-covers-pads warning.
```

P7

```
# P7 digest (stm32-blinky)
Attempt 1: 44/45 connections; U1.8 GND sealed by neighbour escapes; router
refused 0.009mm-margin fix; returned placement_adjust_request (sanctioned).
Backward edge: budget consumed, pre-route restore, U1 cluster +1.0mm east
(placement agent; router's raw suggestions geometrically corrected), place
gate re-PASS. Attempt 2: full chain re-run -> drc_routed PASS 0 err/0 warn,
completion 1.0 (45/45), 207 tracks/20 vias/single GND pour intact.
route_cleanup regressed BOTH attempts (union-find/fill edge, V13) - restored
+ continued without, per design; now flagged unambiguous-bug for hardening.
stitch_impossible advisory FP on track-connected pads noted.
```

P8

```
# P8 digest (stm32-blinky)
verify gate: FAIL a1 (pdn_undecoupled /VIN + 2 pdn_no_bulk FPs) -> both were
CHECK defects: parse_farads multi-token bug fixed + "pdn: false" width-only
opt-out added (schema + tests). PASS a2, all 8 checks 0.
verify-reviewer (fable, fresh): 1 error (J1 polarity silk MISSING - a
requirements promise) + 4 warnings. Fix required silk text -> V17 CLOSED:
add_text/move_text ops built into place_swig/place_edit (idempotent,
independently verified, 9 tests). Fixer applied J1 "5V+"/"GND" legend, SWD
IO/CLK/3V3/GND pin-locked labels, 4 refdes disambiguations -> board fully
DRC-clean (0 violations of ANY kind), check_silk 0, drc_routed re-PASS 0/0.
BOM role drift + stale artifact fixed inline. H4 AUTO-approved; H3 render
folded here per P0 decision. Root-cause note: EasyEDA lib default puts cap
refdes at local (0,-4mm) - 4mm from its own body (library-level fix candidate).
```

P9

```
# P9 digest (stm32-blinky)
dfm a1 FAIL 14: 12 silk slivers (0.003-0.02mm2, EasyEDA body outlines) +
2 copper "gaps". ALL FOUR root causes were CHECKER defects, fixed w/ tests:
(1) dfm silk sliver warn band <0.05mm2 (golden mutant 0.344 stays error);
(2) gerblib FLAT trace caps vs KiCad's circular apertures -> phantom island
splits (2 FP errors) AND w/2 copper-to-edge understatement (FN direction) -
router diagnosed, refused to greenwash, one-line round-cap fix;
(3) bom_cpl ref->LCSC unmappable from S6 parts.json shape -> board_lcsc_map
reads per-footprint LCSC fields (exist since the P5 field fix);
(4) parts.json role drift synced. dfm a2 PASS 0 err/13 warn (non-gating).
BOM complete, 2 CPL rotation corrections (SOT-223, LQFP per jlc_rotations).
```

P10

```
# P10 digest (stm32-blinky)
quote: 5x assembled 2L 50x40 HASL = $18.77 est ($3.75/unit; setup+stencil
dominate; 0 Extended feeders - tantalum came Basic-classed? no: 20 parts all
counted, n_extended 0 per joints calc), authoritative URL in order.json.
order_submit: ready_for_human, all artifacts hashed, API leg exits with
documented prerequisite (V16). H5 recorded auto-approved-for-package;
payment + jlcdfm upload + CPL polarity eyeball (V15) = the 4 human steps.
RUN (a) COMPLETE: brief -> order-ready with 2 design interactions
(P0 batch, H1) + 0 unplanned manual interventions in-run; every break
became a script/prompt/checkpoint fix. M5 bar (<=5) met.
```

Gate attempt history

Timestamp	Gate	Status	Failing/Total
2026-07-27T22:06:05	erc	pass	0/0
2026-07-27T22:31:40	erc	pass	0/0
2026-07-27T23:51:53	place	pass	0/0
2026-07-28T00:18:19	drc_routed	fail	1/1
2026-07-28T00:29:50	drc_routed	pass	0/0
2026-07-28T00:34:28	verify	fail	1/3
2026-07-28T00:34:29	verify	pass	0/0
2026-07-28T01:09:59	drc_routed	pass	0/0
2026-07-28T01:10:00	verify	pass	0/0
2026-07-28T01:15:19	dfm	fail	14/16
2026-07-28T01:25:18	dfm	pass	0/13
2026-08-14T10:05:48	erc	pass	0/0
2026-08-14T10:38:08	place	pass	0/0
2026-08-14T10:38:31	drc_routed	pass	0/0
2026-08-14T10:38:32	verify	pass	0/5
2026-08-14T10:38:35	dfm	pass	0/13

Issues

Id	Phase	Kinds	Severity	Status
1		ldo-cout-stability	error	fixed
2		vdda-decoupling-under-spec	warning	fixed
3		vdd3-bulk-missing	warning	fixed
4		erc-pintypes-vacuous	warning	waived
5		silk-polarity-missing	error	fixed
6		silk-pin-labels-missing	warning	fixed
7		silk-refdes-ambiguity	warning	fixed
8		bom-qty-drift	warning	fixed
9		stale-verify-artifact	warning	fixed

Human checkpoints

Id	Phase	Status	When	Note
1	P2	approved	2026-07-27T17:25:55	Architecture approved as recommended (SS34 kept)
2	P4	approved	2026-07-27T22:31:41	AUTO-approved per user directive. Schematic PDF reports/schematic.pdf; review 1 err + 3 warn -> err+2 warn fixed pre-board, 1 warn waived (erc-waivers.md); ERC 0/0; audit pass
3	P6	skipped	2026-07-27T23:51:54	Folded into H4 per user-approved P0 decision; render at reports/place/render_final/
4	P8	approved	2026-07-28T01:10:01	AUTO-approved per user directive. verify 8/8 clean; drc_routed 0/0; reviewer 1 err + 4 warn: err (J1 polarity silk) + 4 warn ALL FIXED via new V17 text ops + parts.json role sync + fresh artifact; remaining waiver = minor refdes margin (C1 2.2mm own vs 3.0mm R1). Renders: reports/render_labeled/ + reports/place/render_final/ (H3 fold)

Id	Phase	Status	When	Note
5	P10	approved	2026-07-28T01:26:02	AUTO-approved package per user directive: order.json ready_for_human, zip sha b013238b, est 8.77/5 assembled. PAYMENT + JLC uploads (jlcdfm, quote, CPL polarity preview = V15) remain the human's 4 listed steps - never automated

10 Artifact Index

File	Size	Note
`state.json`	33,592 B	
`kicad/stm32-blinky.kicad_pcb`	154,571 B	
`kicad/stm32-blinky.kicad_sch`	114,806 B	
`fab/stm32-blinky_gerbers.zip`	21,472 B	sha256 9a5604b0ab151805...
`fab/order.json`	2,354 B	
`fab/BOM.csv`	632 B	
`fab/CPL.csv`	703 B	
`brief/brief.md`	384 B	
`requirements.md`	4,281 B	
`architecture/blocks.md`	4,299 B	
`architecture/constraints.json`	1,046 B	
`architecture/decisions.md`	2,458 B	
`architecture/power_tree.md`	3,090 B	
`architecture/sheets.md`	3,742 B	
`architecture/stackup.md`	1,577 B	
`reports/board_init.json`	9,575 B	
`reports/bom_cpl.json`	12,275 B	
`reports/dfm_release.json`	10,600 B	
`reports/erc-waivers.md`	725 B	
`reports/fab_export.json`	2,330 B	
`reports/fp_C32713268_HDR2P.overlay.svg`	1,135 B	
`reports/fp_C32713270_HDR4P.overlay.svg`	1,863 B	
`reports/fp_C6186_SOT223.overlay.svg`	1,887 B	
`reports/fp_C8020_CASEB.overlay.svg`	1,132 B	
`reports/fp_C84256_LED0805.overlay.svg`	1,115 B	
`reports/fp_C8734_LQFP48.overlay.svg`	18,048 B	
`reports/fp_verify_C32713268.json`	917 B	
`reports/fp_verify_C32713270.json`	911 B	
`reports/fp_verify_C6186.json`	1,143 B	
`reports/fp_verify_C8020.json`	1,004 B	
`reports/fp_verify_C84256.json`	844 B	
`reports/fp_verify_C8734.json`	1,749 B	
`reports/gate-dfm.json`	1,086 B	
`reports/gate-drc_routed-a1.json`	1,198 B	
`reports/gate-drc_routed.json`	407 B	
`reports/gate-erc.json`	400 B	
`reports/gate-place.json`	898 B	
`reports/gate-verify.json`	1,475 B	
`reports/lib_pull.json`	5,129 B	
`reports/lib_pull_delta1.json`	1,210 B	
`reports/review-board.json`	3,084 B	
`reports/review-board.md`	10,555 B	
`reports/review-schematic.json`	2,008 B	
`reports/review-schematic.md`	7,802 B	

File	Size	Note
reports/rules_gen.json	1,020 B	
reports/schematic.pdf	80,389 B	
reports/top.net	47,736 B	
reports/verify_all.json	2,147 B	
reports/verify_release.json	7,364 B	